

VISION Resistivity
Dual Frequency
Recorded Mode

Schlumberger

Country: Norway

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

Longitude: 3° 26' 36.13" E

Block: 31/5

UWID:

Rig Name:

Rig Type:

West Hercules

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 12 1/4 in. pilot

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 16-Dec-2019 -- 17-Dec-2019

Log Interval: 906.00(m) -- 1907.00(m)

Index Types: Measured Depth

Index Scales: 1:200

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

Other Services:

Direction & Inclination (Surveys)

Directional Drilling

Annular Pressure While Drilling

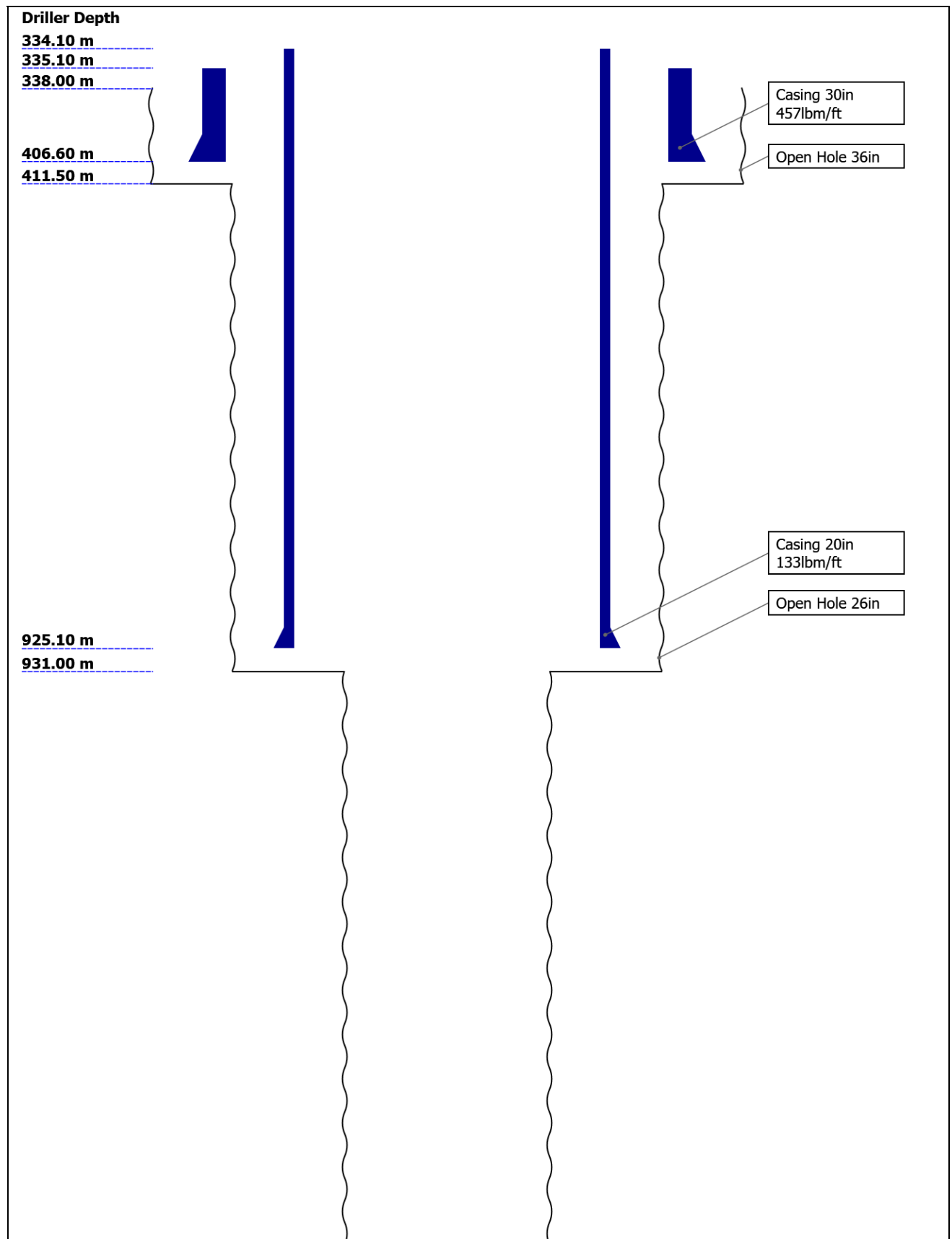
Disclaimer

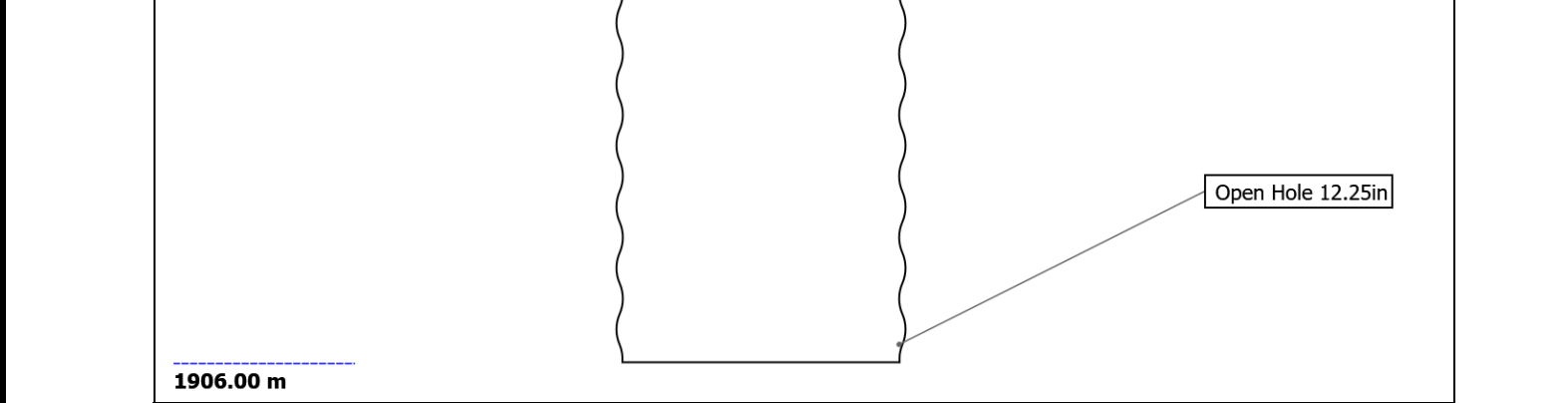
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

1. Header
2. Disclaimer
3. Contents
4. Well Sketch
5. Borehole Size/Casing/Tubing Record
6. Operational Run Summary
7. Borehole Fluids
8. Remarks and Equipment Summary
9. Run 5 VISION Resistivity - Dual Frequency
 - 9.1 Integration Summary
 - 9.2 Software Version
 - 9.3 Composite Summary
 - 9.4 Log (Resistivity - Dual Frequency)
 - 9.5 Parameter Listing
10. Calibration Report
11. Tail

Well Sketch





Borehole Size/Casing Record						
Bit						
Bit Size (in)	36	26	12.25			
Top Driller (m)	338	411.5	931			
Bottom Driller (m)	411.5	931	1906			
Casing						
Size (in)	30	20				
Weight (lbm/ft)	457	133				
Inner Diameter (in)	27.067	18.73				
Grade	X56	N80				
Top Driller (m)	335.1	334.1				
Bottom Driller (m)	406.6	925.1				

Operational Run Summary						
Parameter (unit)	Run 5					
Date Log Started	16-Dec-2019					
Time Log Started	08:19:22					
Date Log Finished	17-Dec-2019					
Time Log Finished	14:10:12					
Bit Size (in)	12.250					
Bit Start Depth (m)	931.00					
Bit Stop Depth (m)	1906.00					
Top Log Interval (m)	925.80					
Bottom Log Interval (m)	1896.80					
Max Hole Deviation (deg)	1.02					
Azimuth of Max Deviation (deg)	139.83					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	S.Vikra					
Service Order Number	19SCA0085					

Borehole Fluids

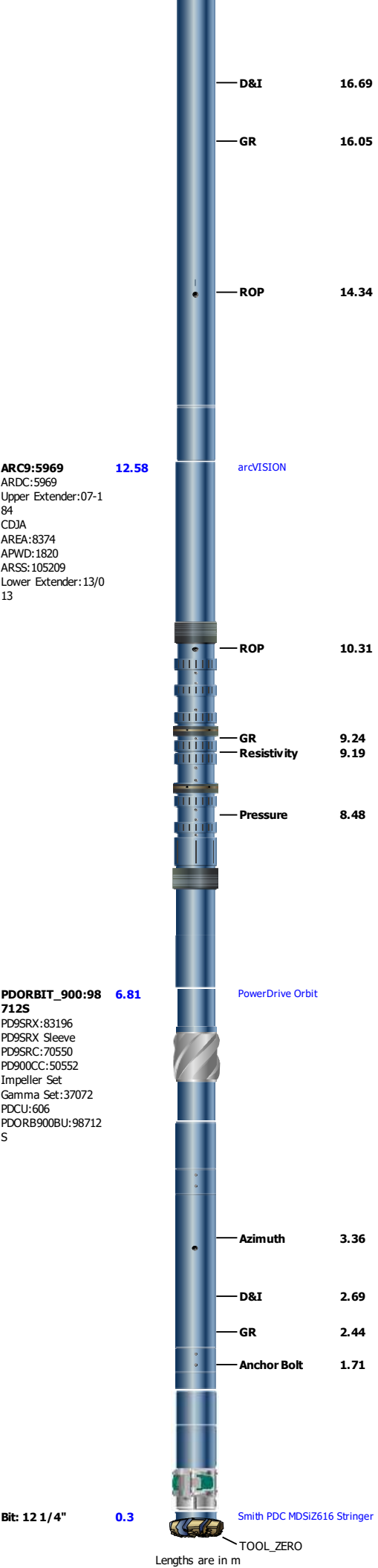
Parameter(unit)	Run 5					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	34					
Source of Sample	Active Tank					
Salinity (ppm)	Zoned					
Density (g/cm3)	1.25					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.08 @ 17.78					
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 18.6					
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 20					
RM @ BHT (ohm.m@degC)	0.05 @ 34					
RMF @ BHT (ohm.m@degC)	0.05 @ 34					
RMC @ BHT (ohm.m@degC)	0.07 @ 34					
Total Solid (%)						
High Gravity Solids (%)						

Run 5

Parameter	Value	Start
Salinity	114390.2	12/16/2019 8:19:22 AM
Salinity	114390.2	12/16/2019 11:56:31 AM

Remarks and Equipment Summary

Run 5: Toolstring			Run 5: Remarks
Equip name Stabilizer:MWS 37 32	Length 24.59	MP name 12 1/8 in Non-Mag Stab	All depths referenced to driller's depth. See EOWR for depth tracking details.
			All data from tool memory. All data acquired while drilling
			Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
X/O: 9 1/2"[2]:XO S-947	22.09	X-Over	Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
X/O: 9 1/2"[1]:71 539	21.26	X-Over	Run 5 and 12 1/4 in. pilot section TD at 1906 m.
TELE900:G5093 MSSU900:71539 Upper Extender:26-0 2-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PMEA:3708 MTA:2526 MTK900:008/298895 -2142 MSSD900:973247-1 Lower Extender:12/1 42	20.62	TeleScope	



Run 5

VISION Resistivity - Dual Frequency

Software Version	
------------------	--

Acquisition System	Version
Maxwell 2019.2	9.2.113335.3100

Computation	Description	Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100
ARC9GammaRayComputation	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100

Tool Interface	System Version	Loaded Version
HSPM	2019.0c.001	9.2.113335.3100

Tool Elements	Description	Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar	9.2.113335.3100	V9.7B

Pass Summary	
--------------	--

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 5	Drilling	Down	22.96 m	1906.17 m	16-Dec-2019 8:19:22 AM	17-Dec-2019 2:10:12 PM	Yes

All depths are referenced to toolstring zero

Log	Company:Equinor Well:31/5-7 Run 5: Drilling:S099
-----	--

Company:Equinor Well:31/5-7

Run 5: Drilling:S099

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 20-Dec-2019 14:08:37

-|TICK ARC_RES - Resistivity Tick Marks ARC9 RM

└ TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

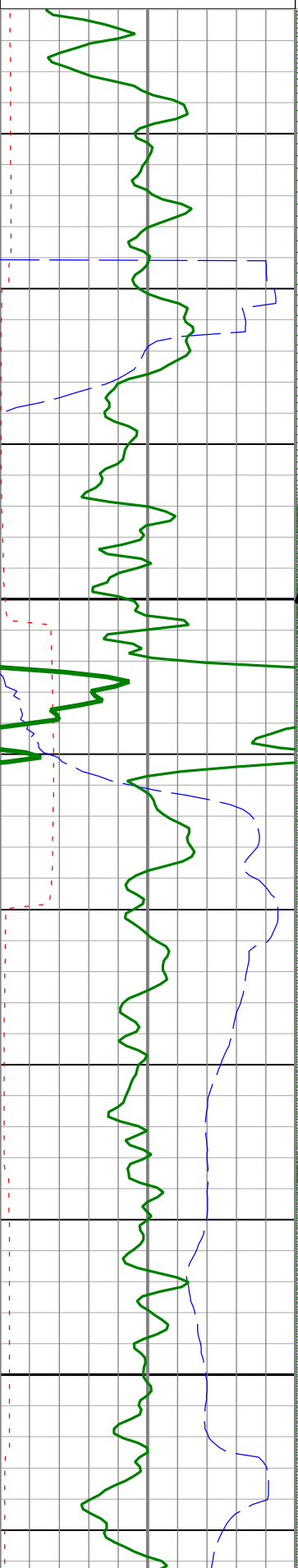
Resistivity Time After Bit (ARTM) ARC9 0h10 Rate of penetration averaged over	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2ohm.m200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2ohm.m200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2ohm.m200

Rate of penetration averaged over
the last 5 ft (1.5 m) (ROP5) RT

200 m/h 0

Gamma Ray (GR_ARC) ARC9 RM

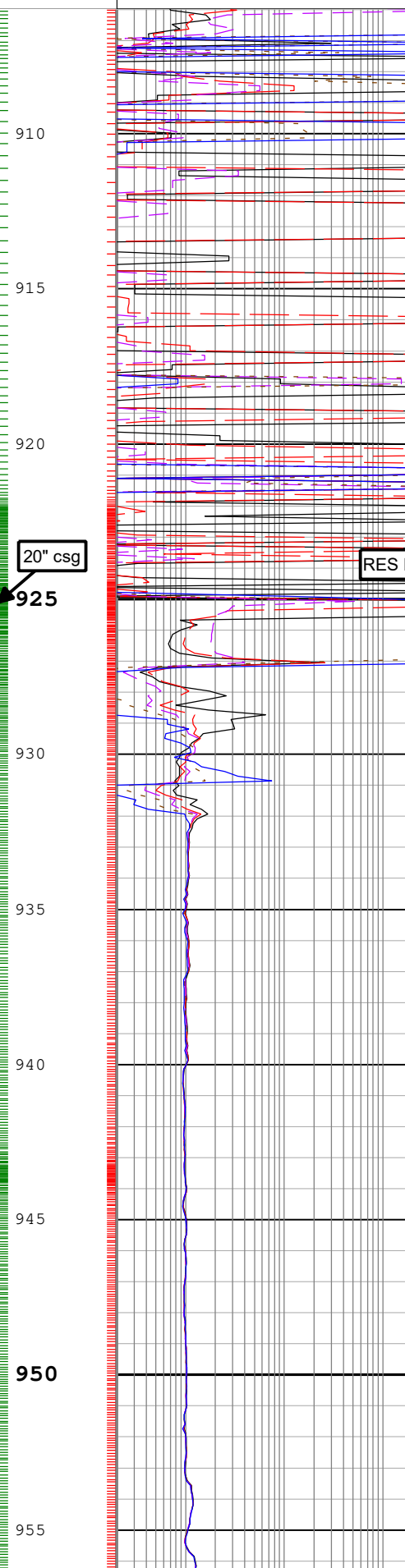
0 gAPI 150



0.2 ohm.m 200

Phase Shift Resistivity 16 inch
Spacing at 2 MHz, Environmentally
Corrected. (P16H) ARC9 RM

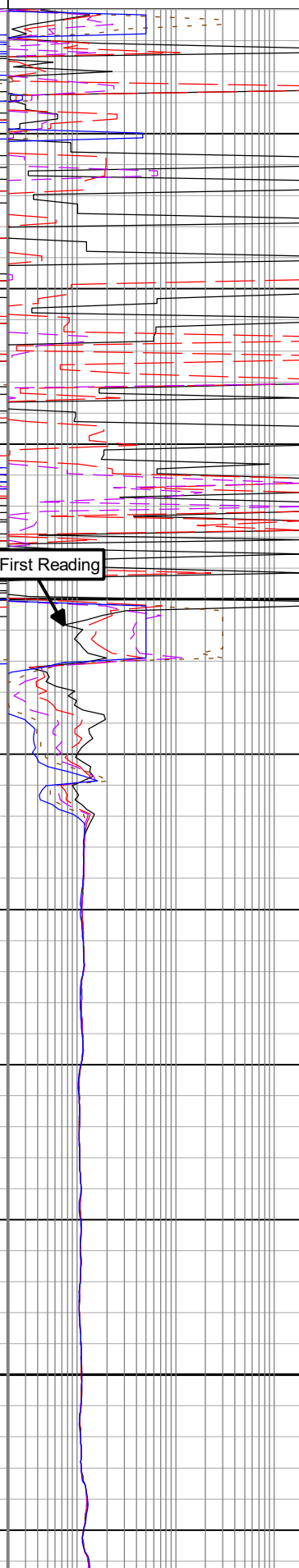
0.2 ohm.m 200



0.2 ohm.m 200

Attenuation Resistivity 16 inch
Spacing at 2 MHz, Environmentally
Corrected. (A16H) ARC9 RM

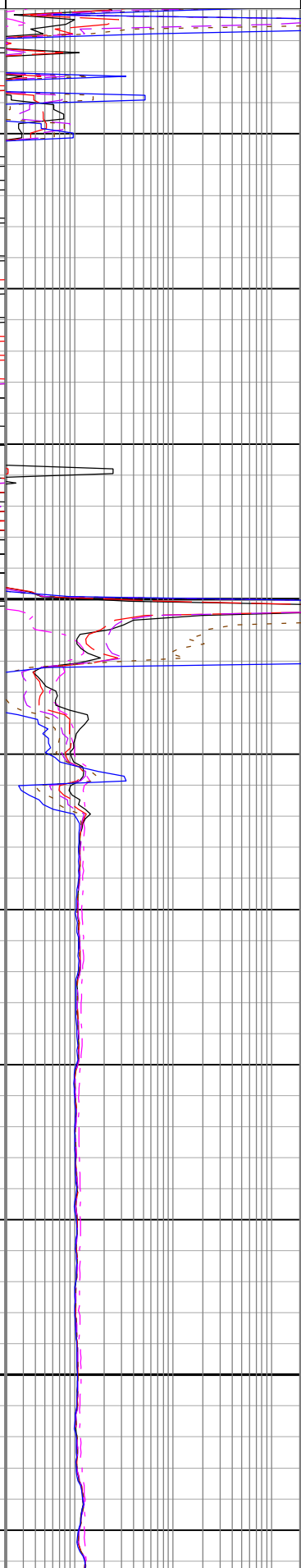
0.2 ohm.m 200

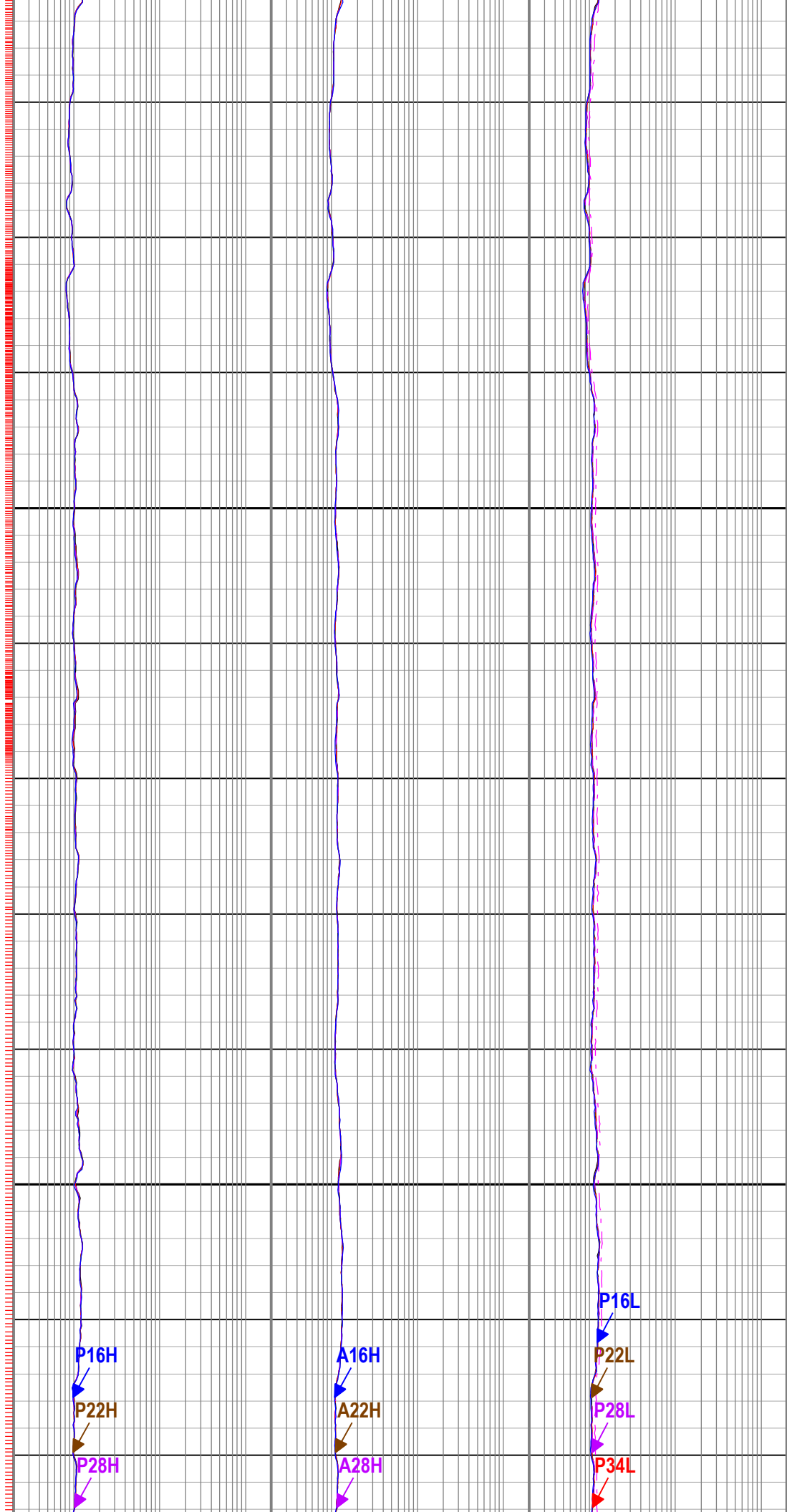
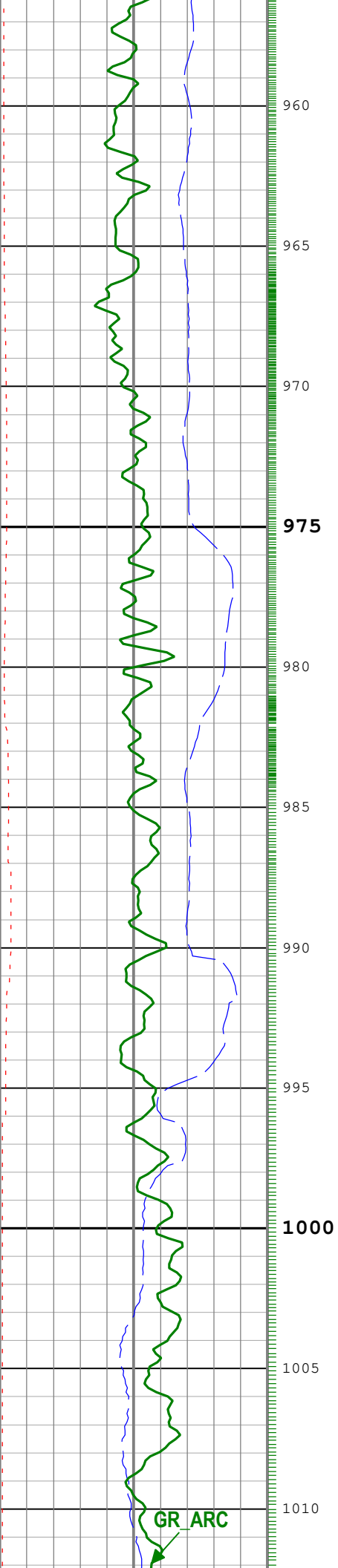


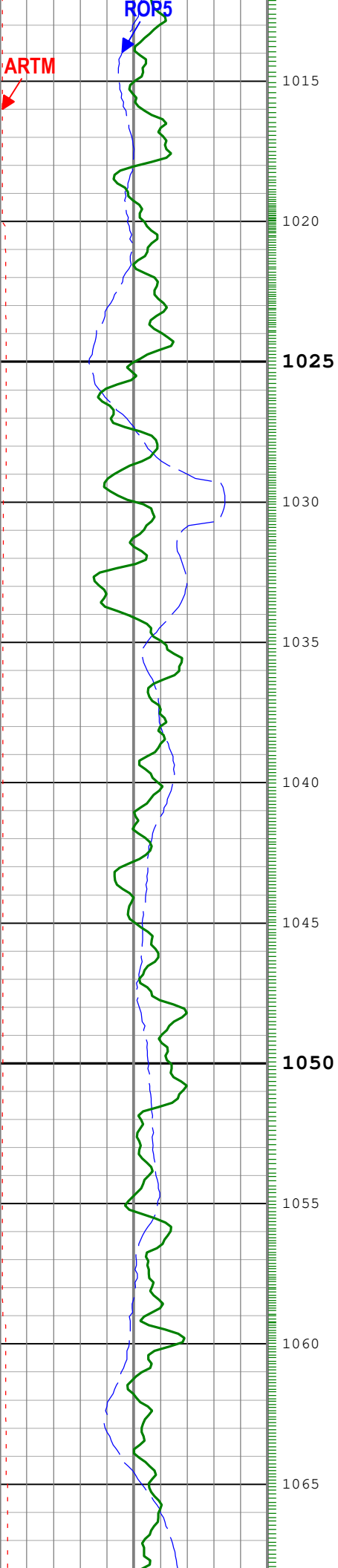
0.2 ohm.m 200

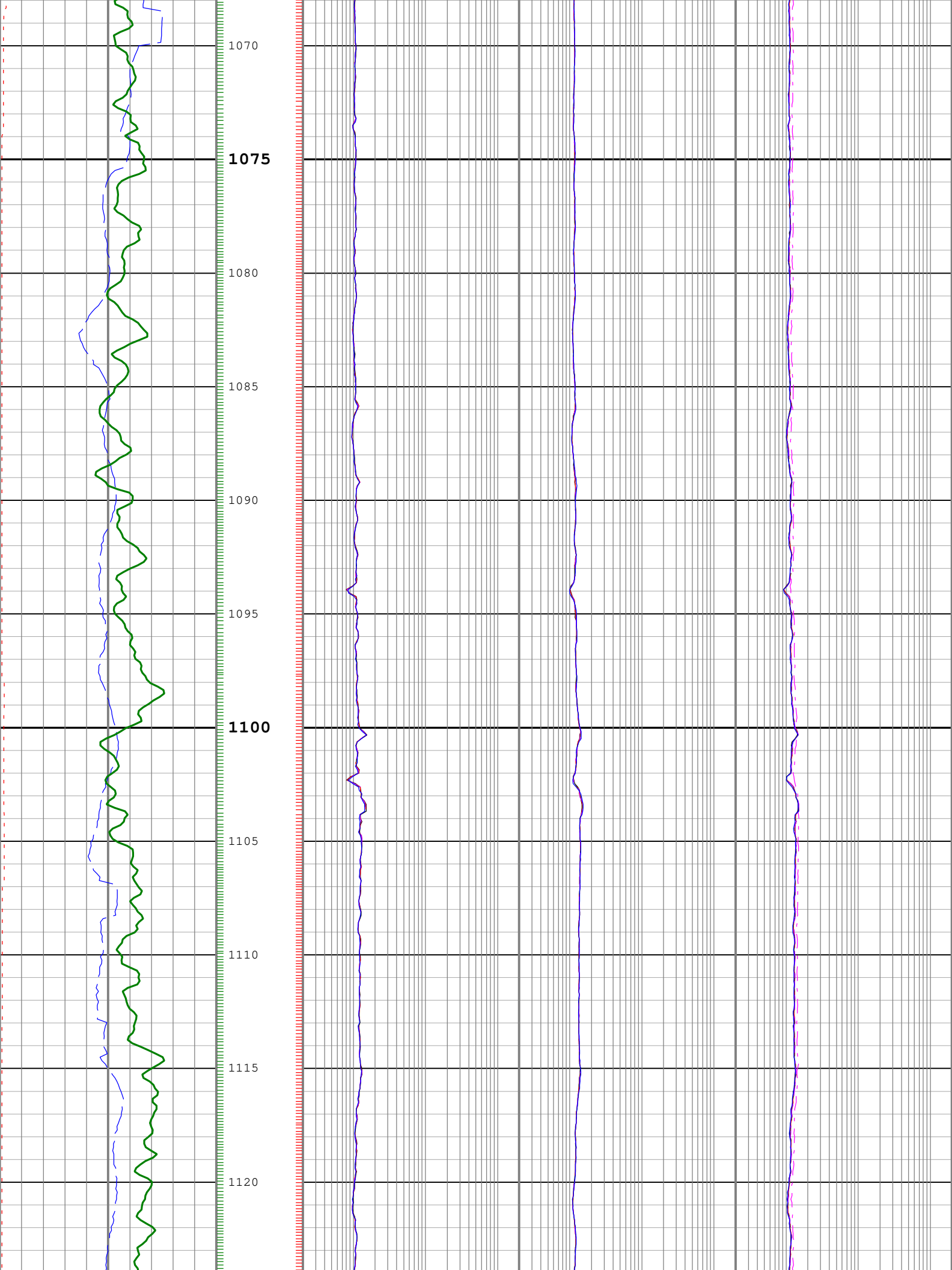
Phase Shift Resistivity 16 inch
Spacing at 400 KHz,
Environmentally Corrected. (P16L)
ARC9 RM

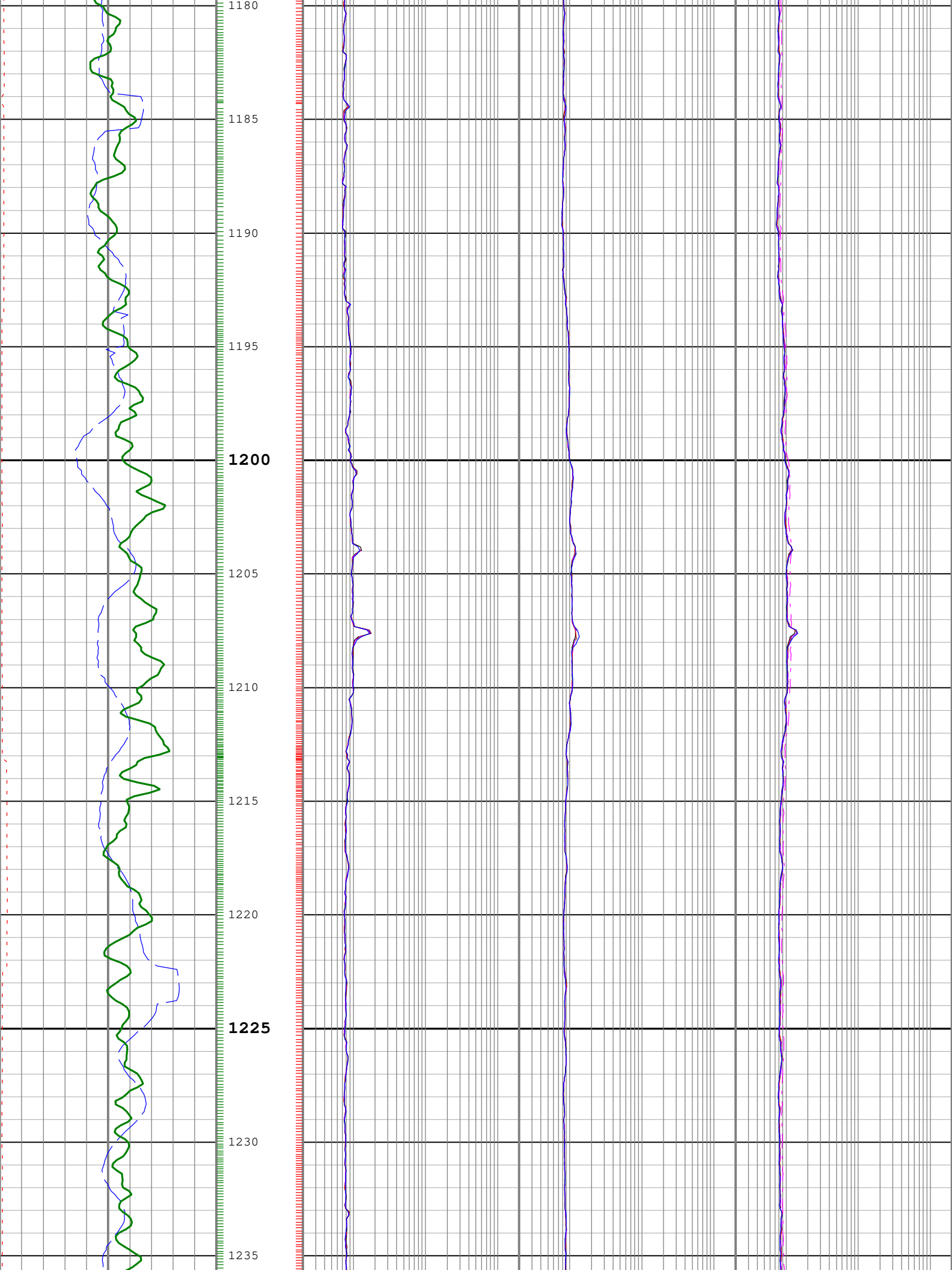
0.2 ohm.m 200

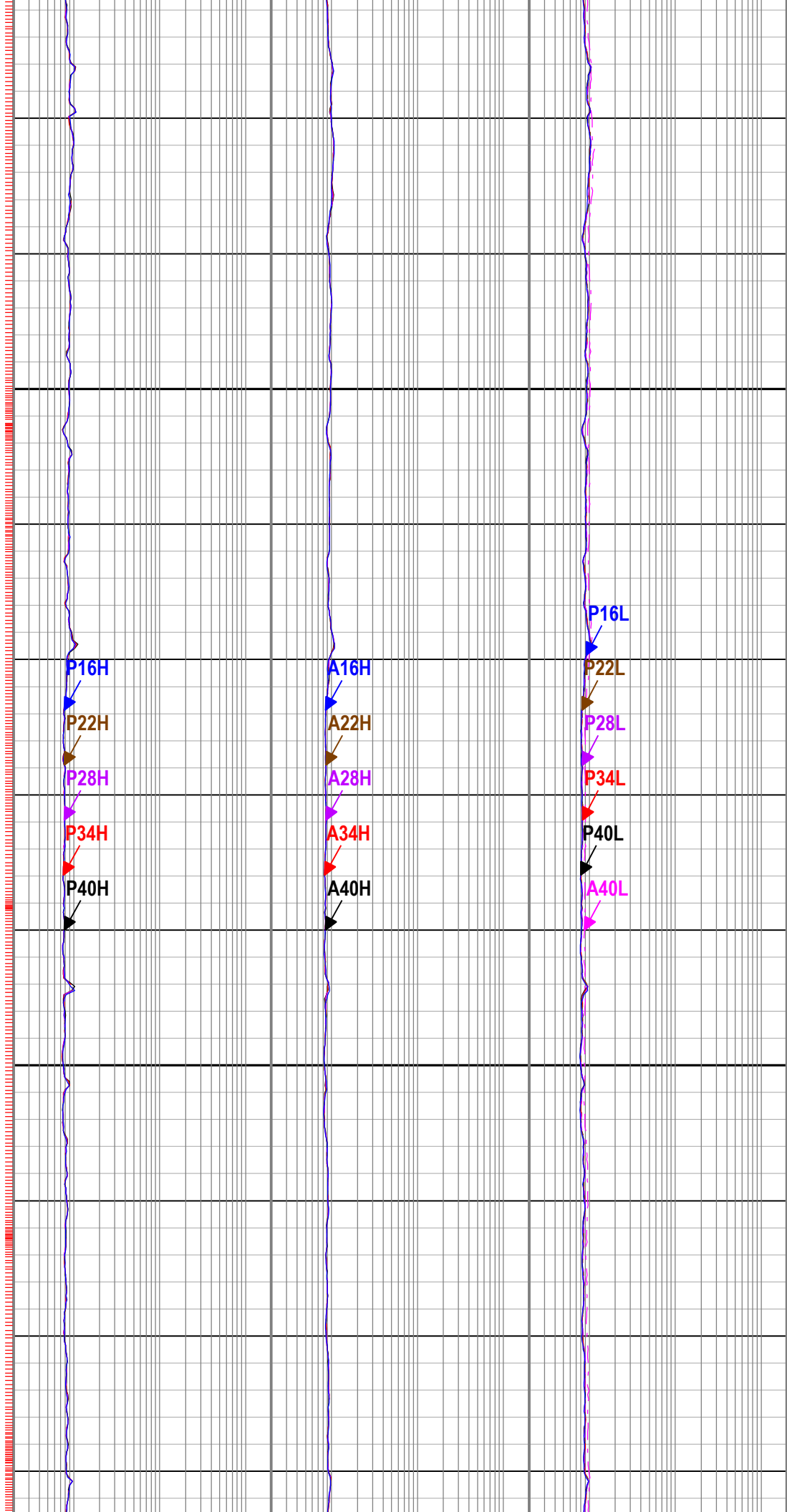
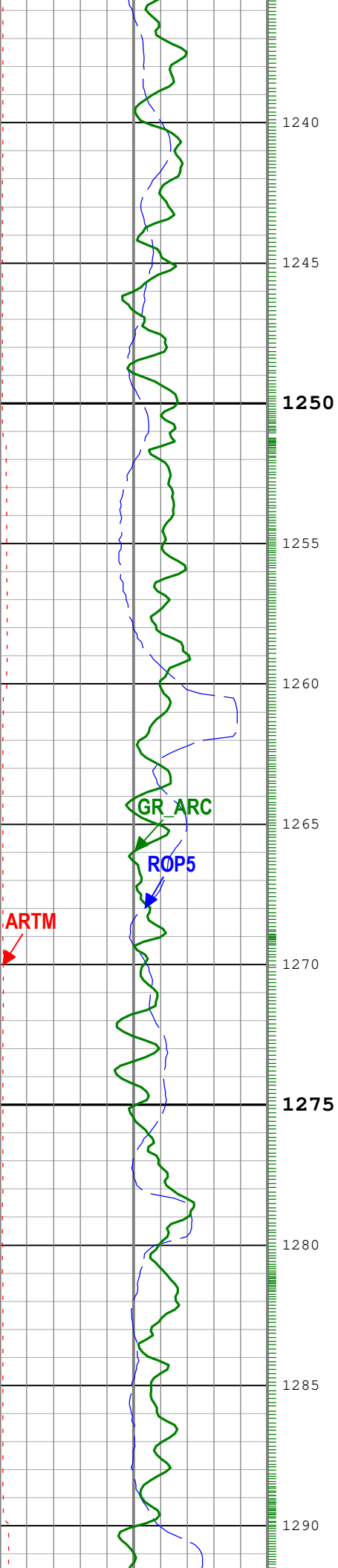


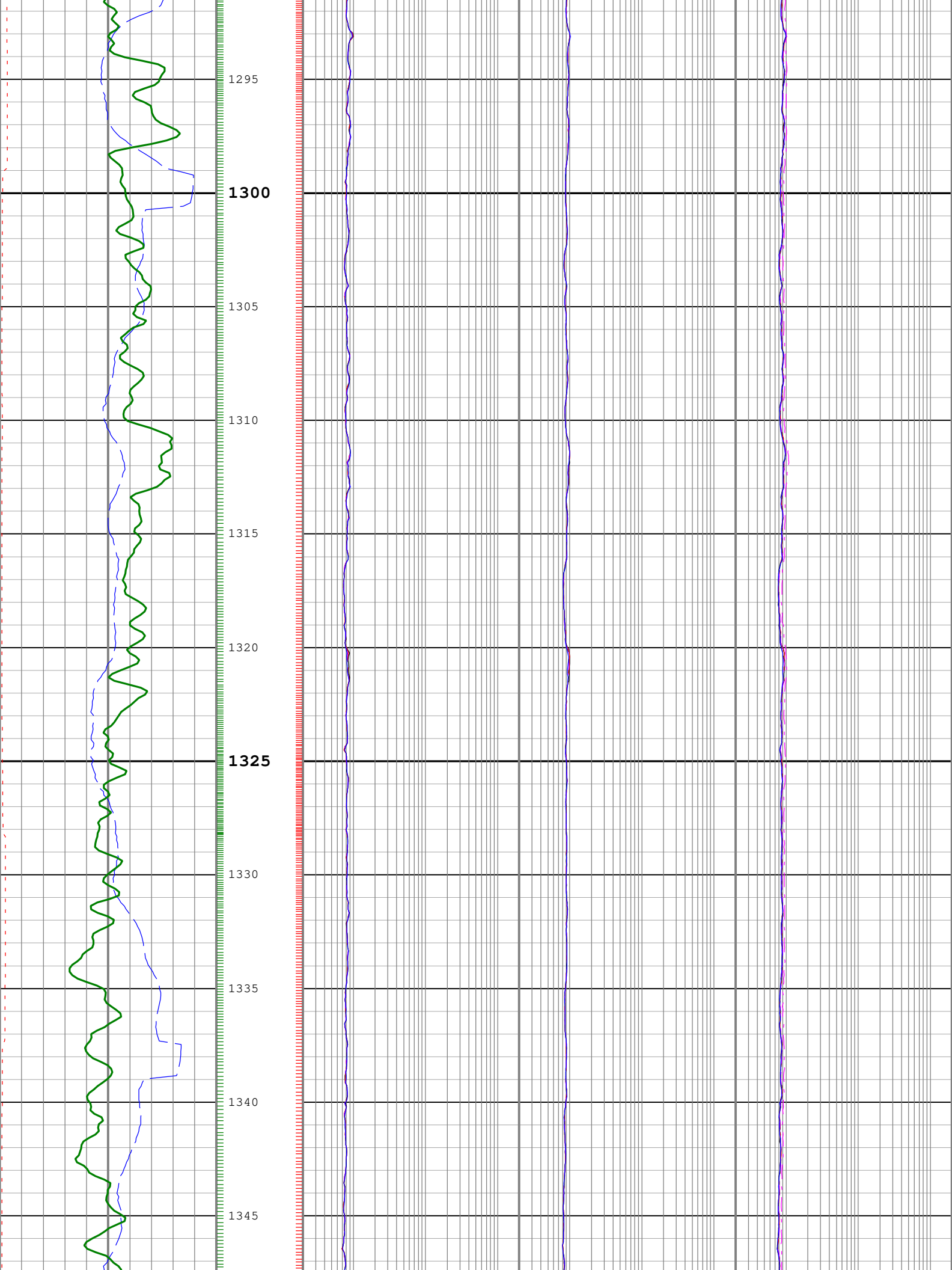


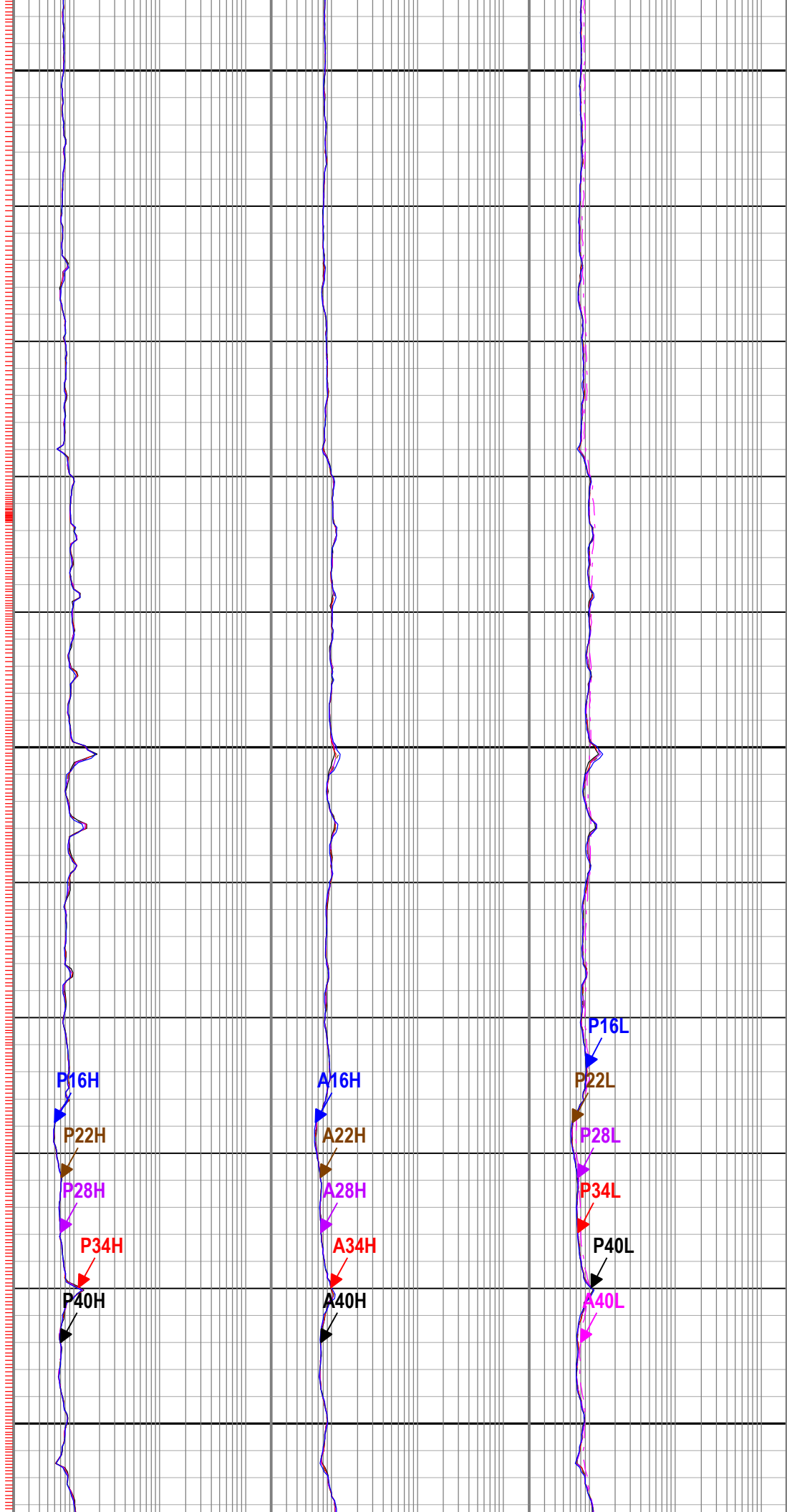
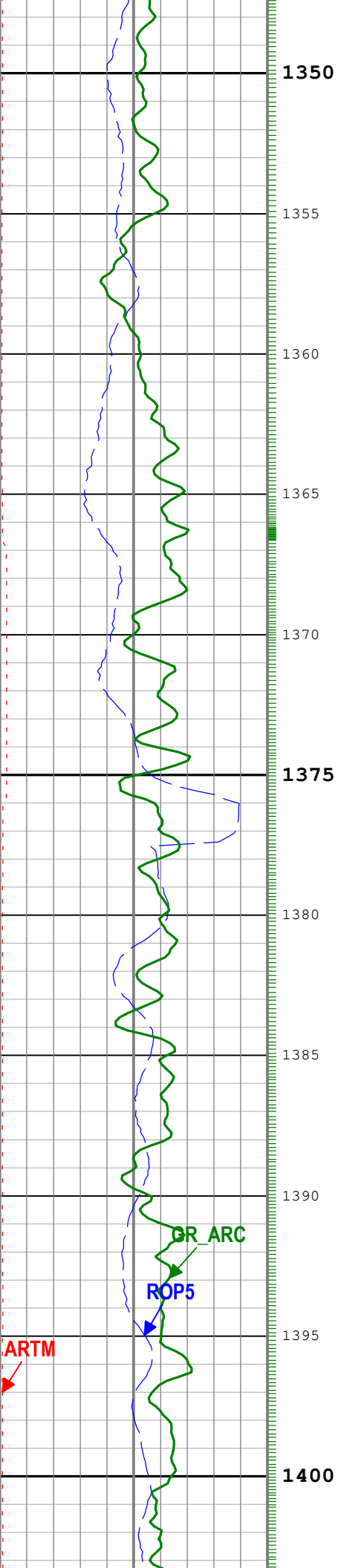


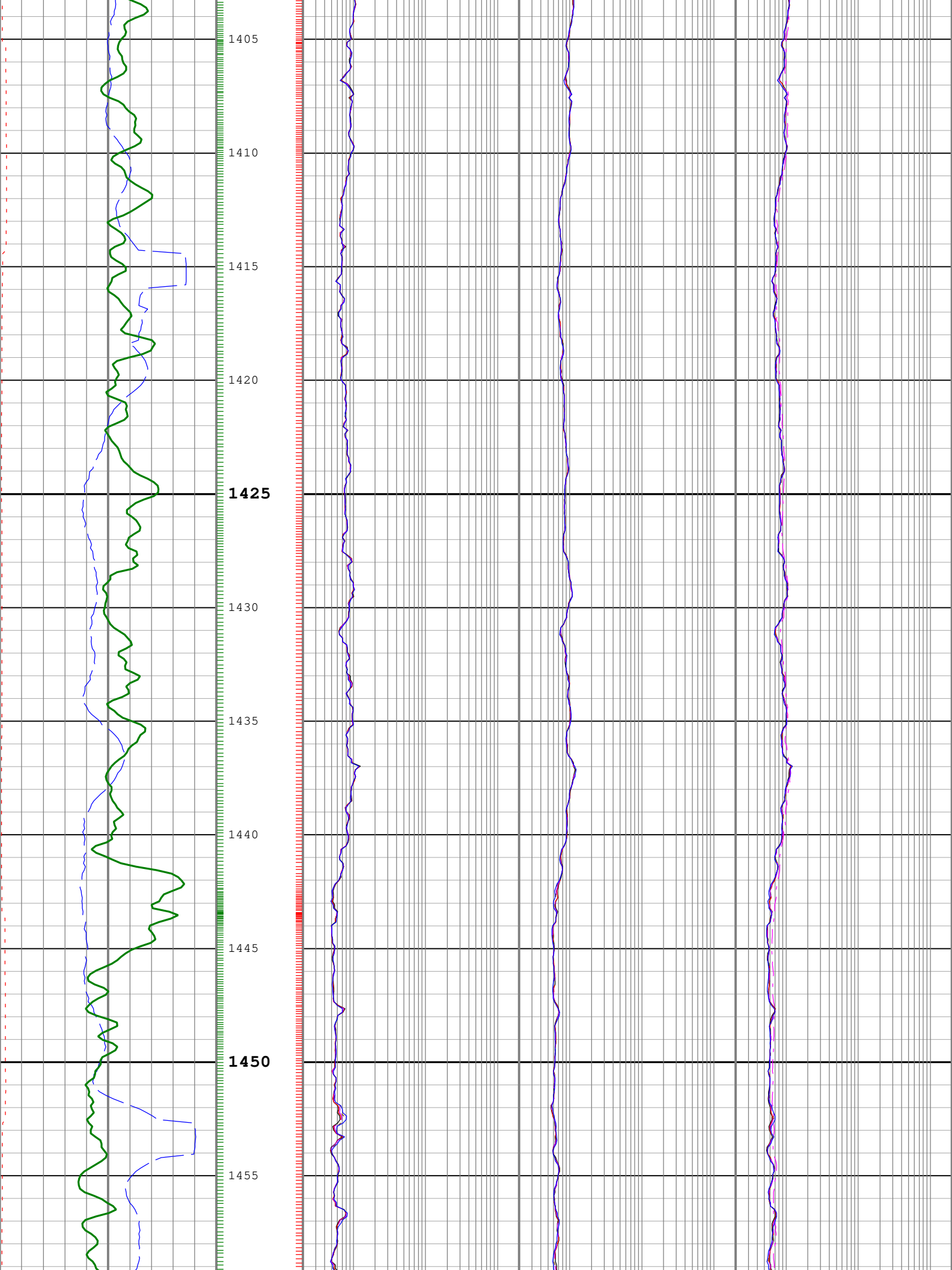


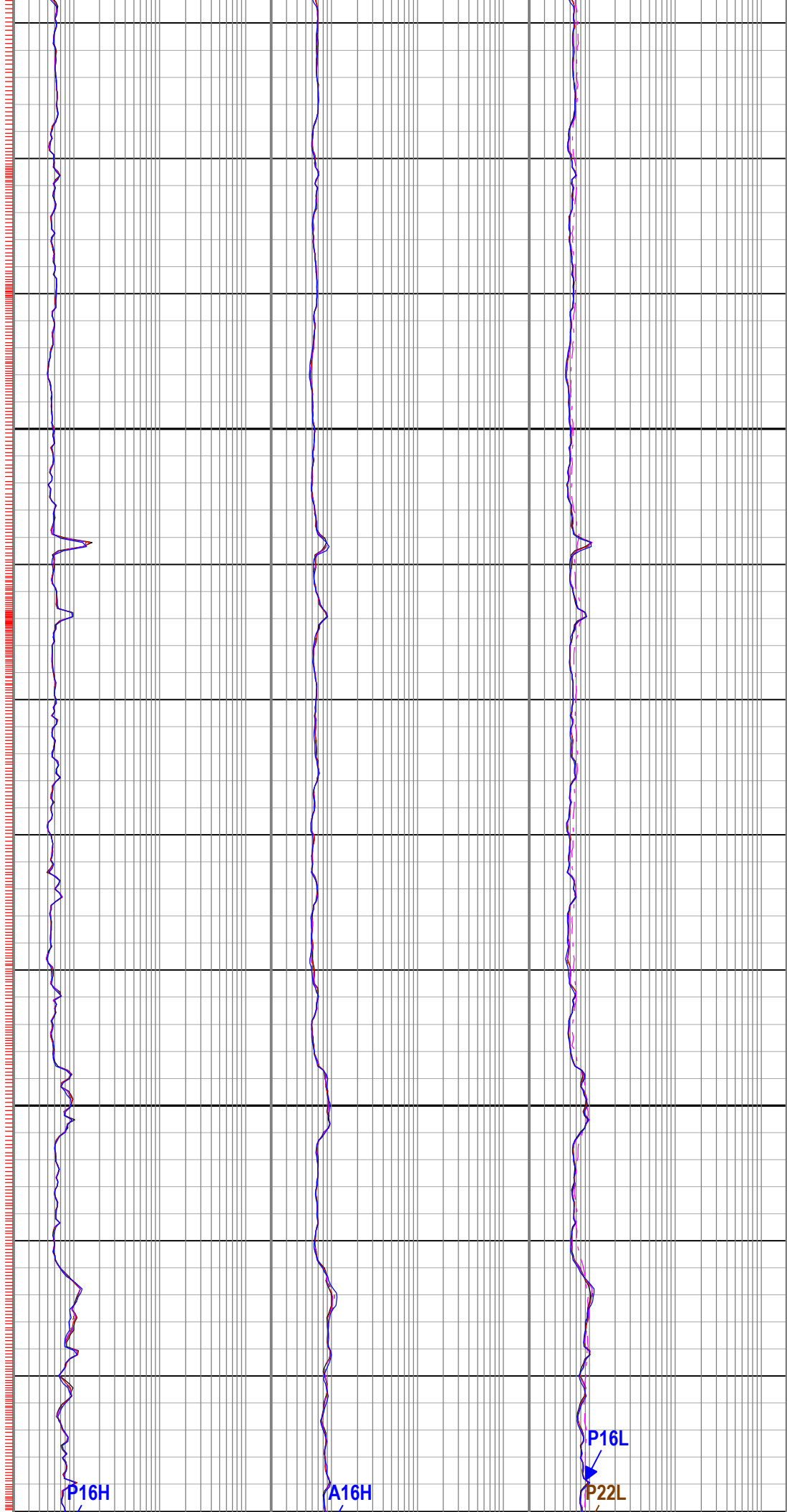
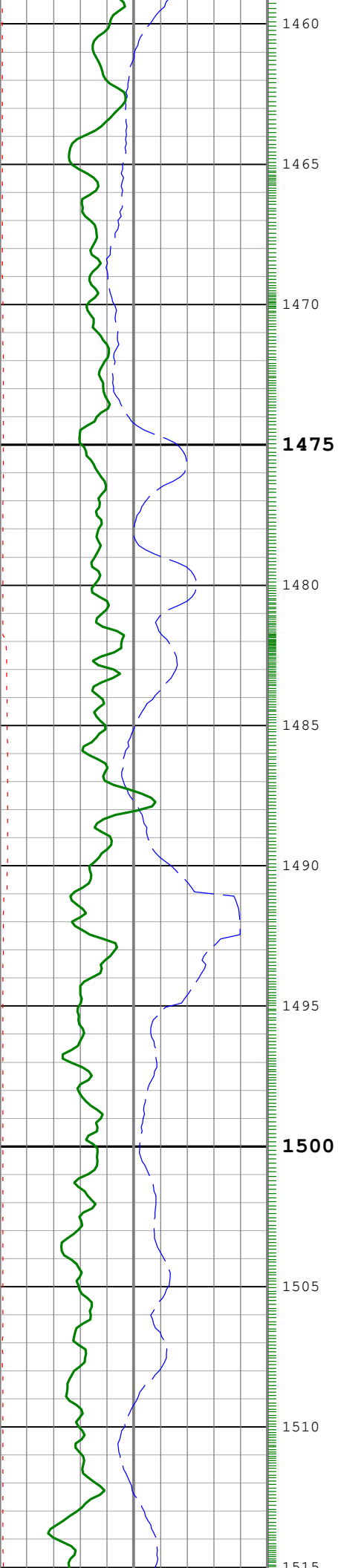


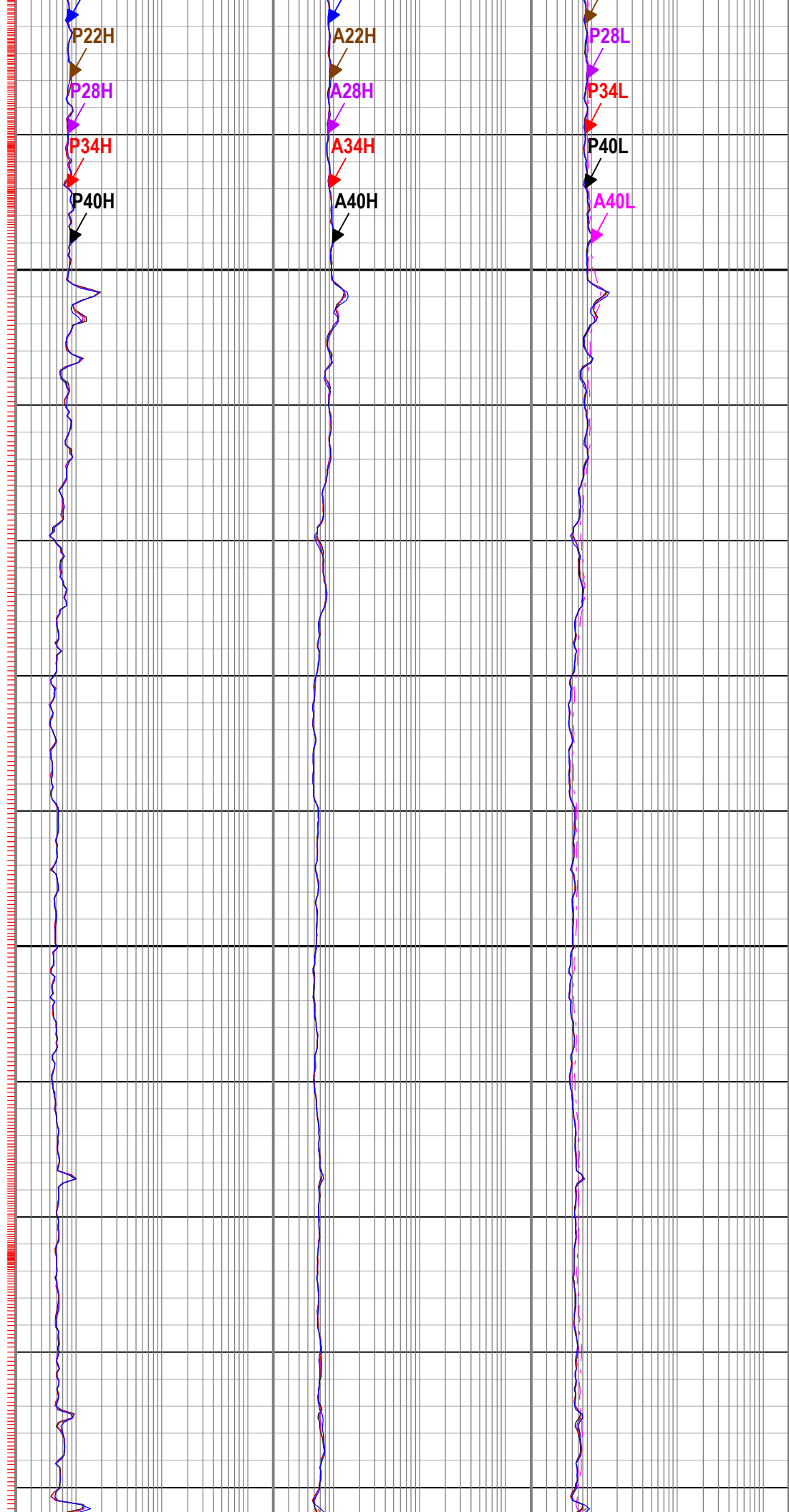
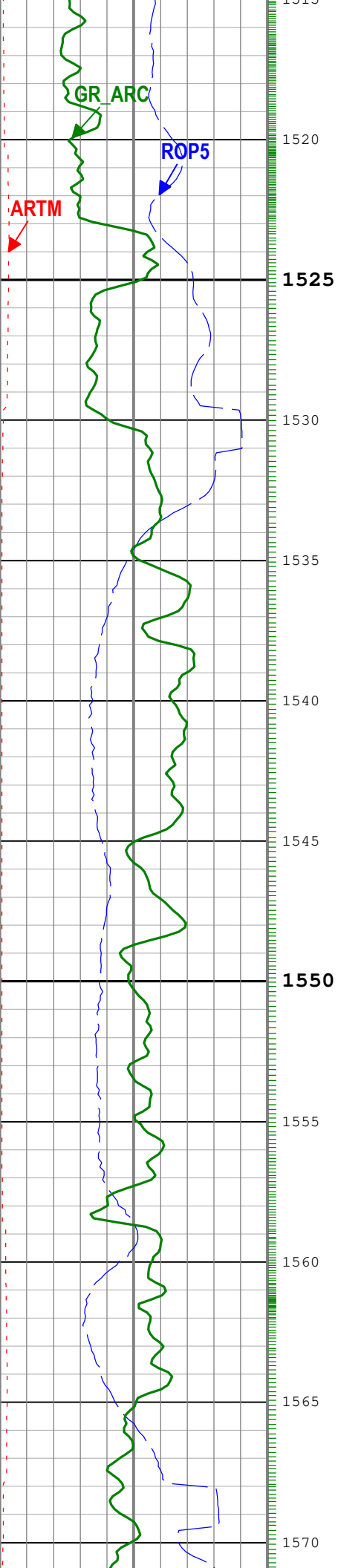


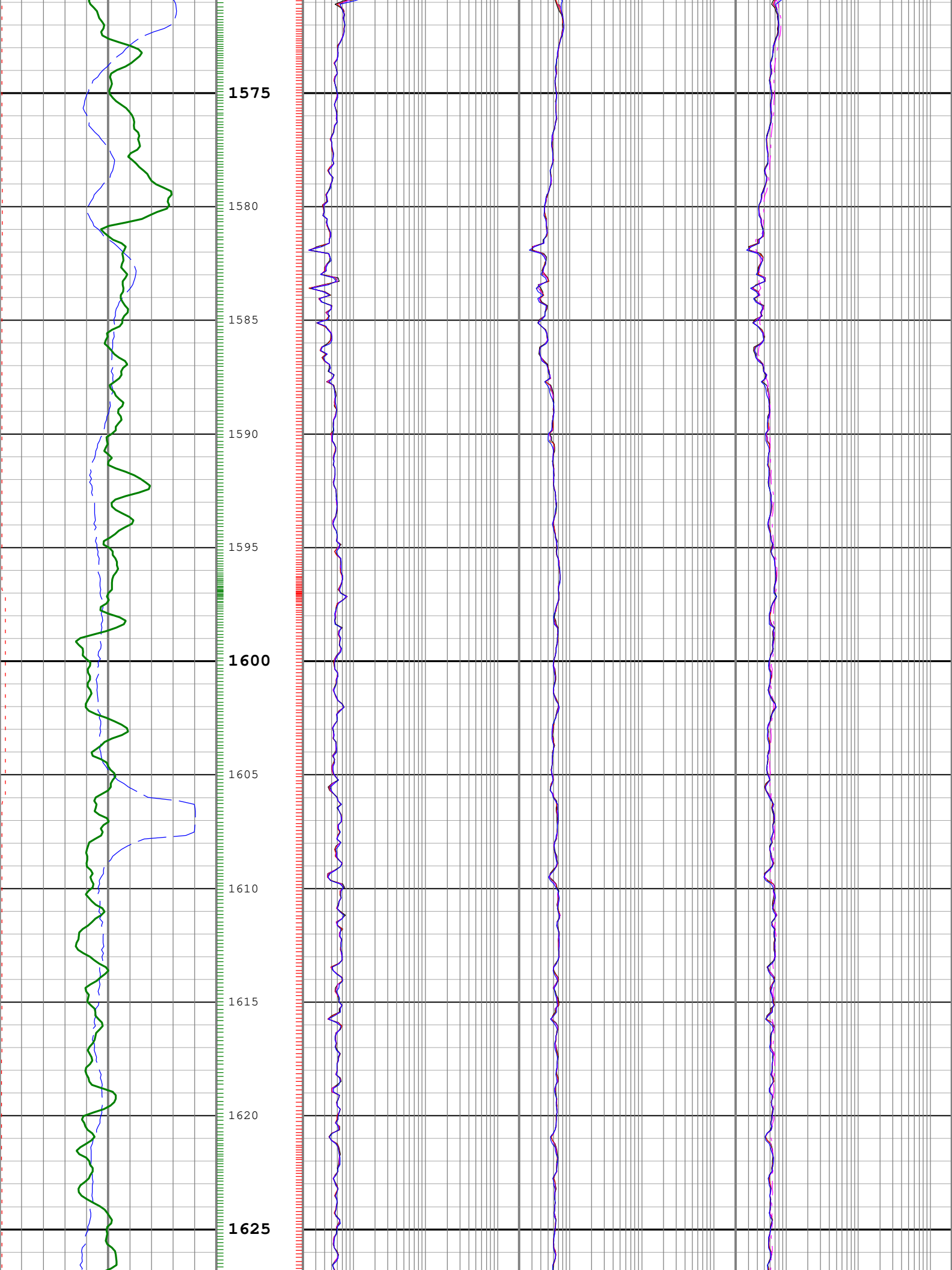


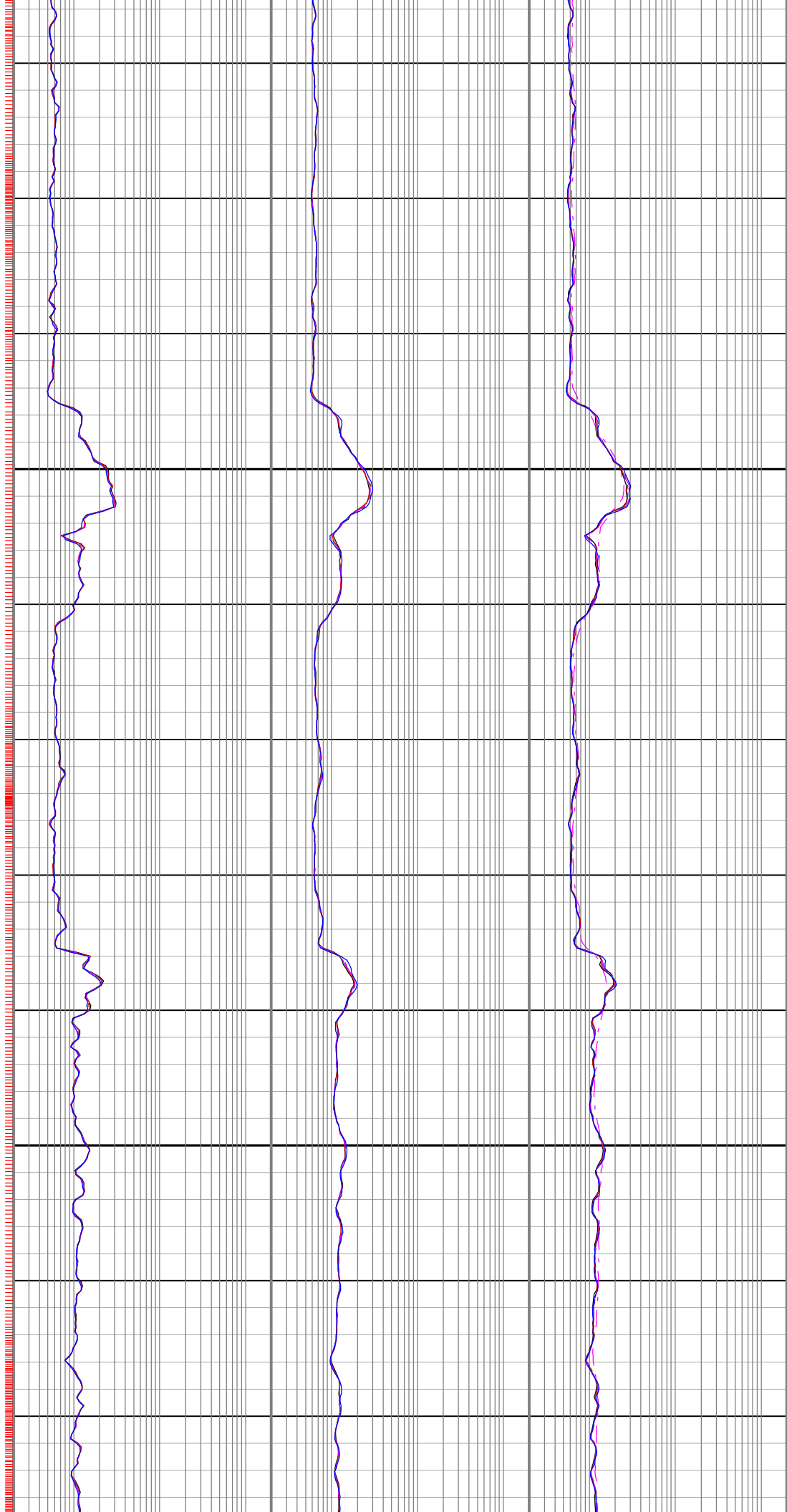
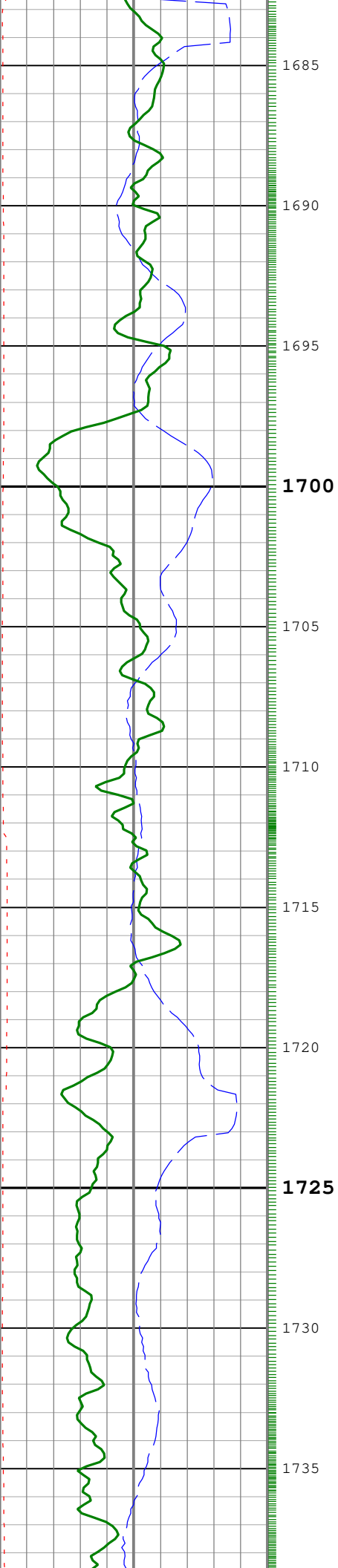


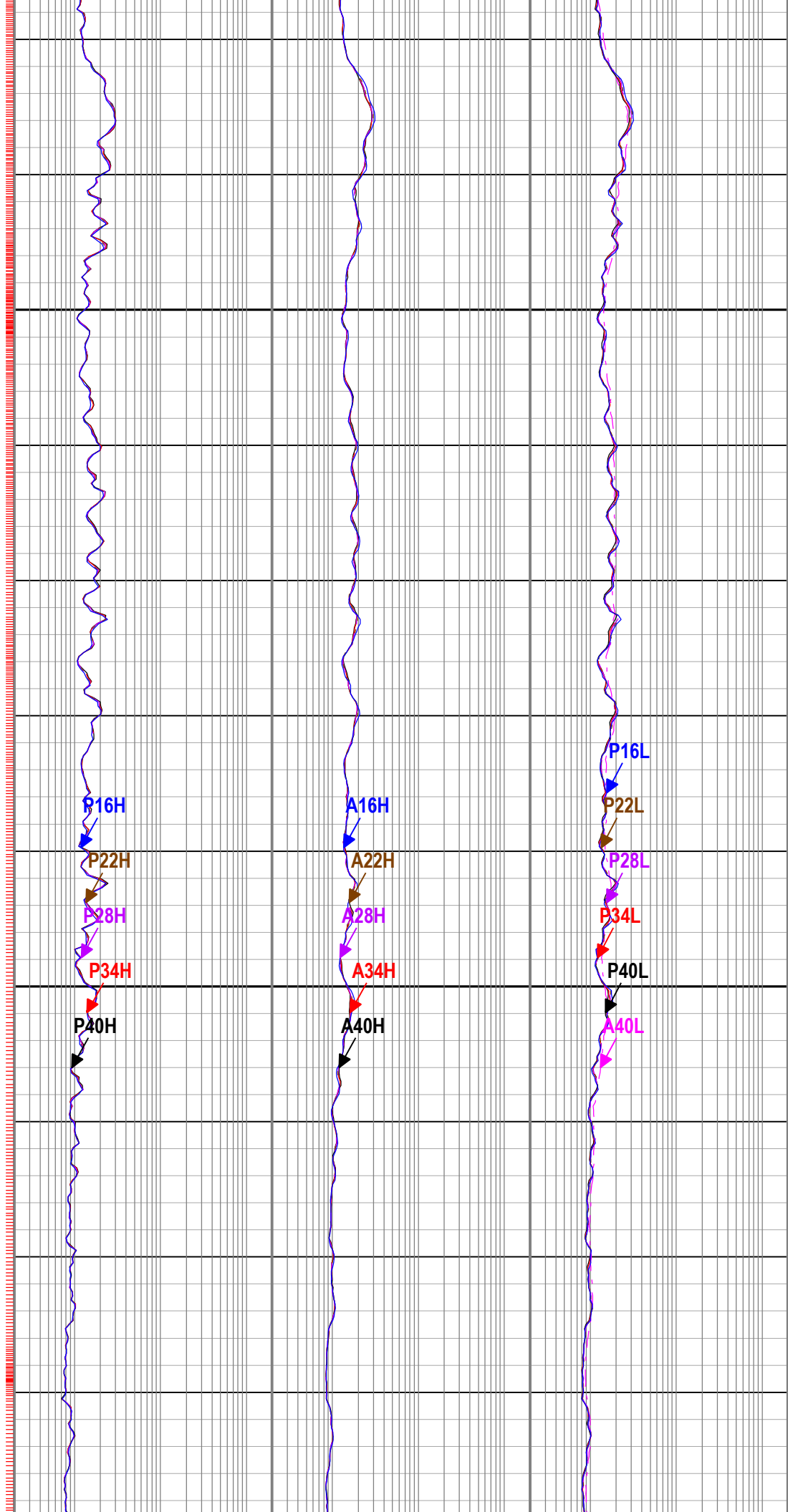
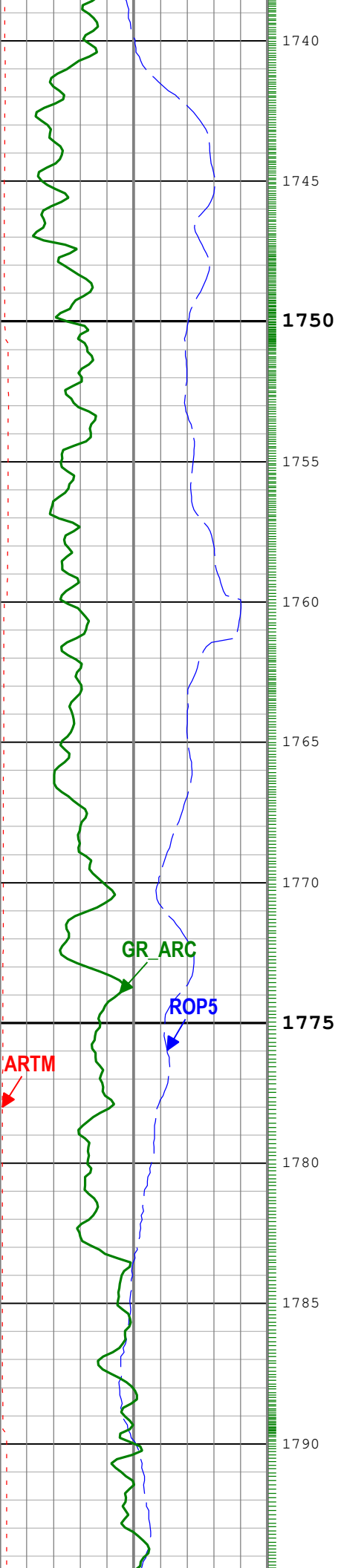


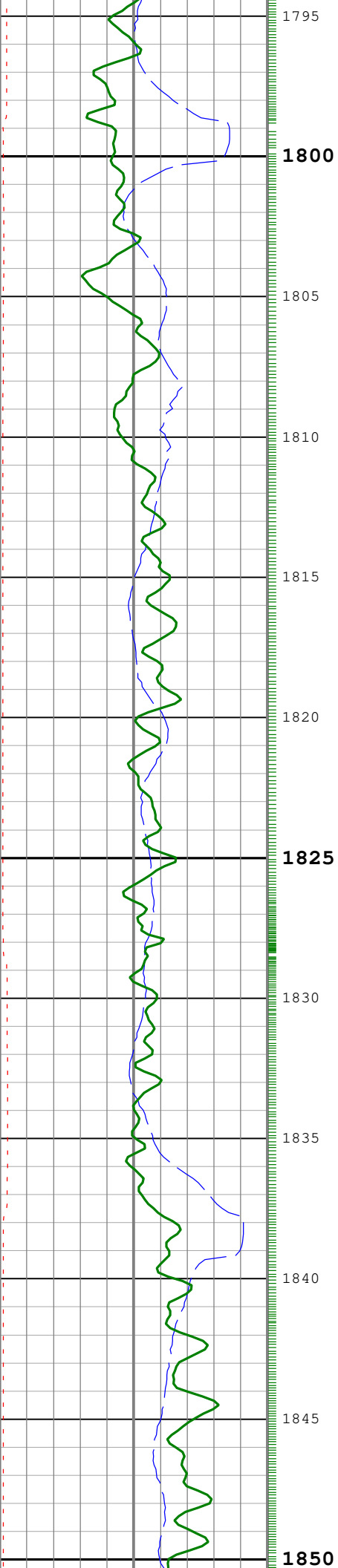


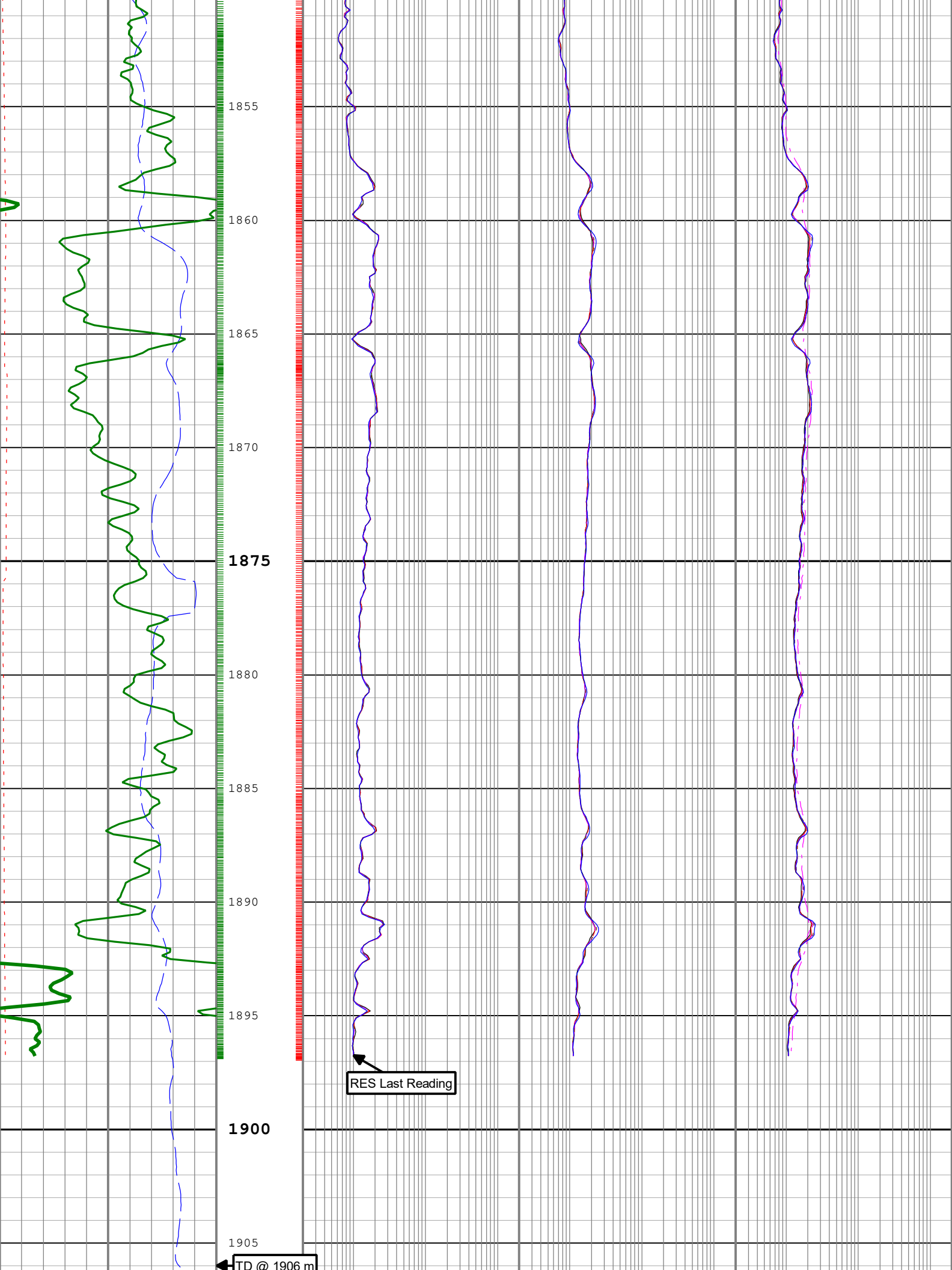












Resistivity Time After Bit (ARTM) ARC9 0 h 10 Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200 m/h 0 Gamma Ray (GR_ARC) ARC9 RM 0 gAPI 150		Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM 0.2 ohm.m 200
		Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM 0.2 ohm.m 200
		Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2 ohm.m 200
		Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2 ohm.m 200
		Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2 ohm.m 200
				Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM 0.2 ohm.m 200

┆TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

┆TICK_ARC_RES - Resistivity Tick Marks ARC9 RM

Description:
Format: Log (Resistivity - Dual Frequency)
Index Scale: 1:200
Index Unit: m
Index Type: Measured Depth
Creation Date: 20-Dec-2019 14:08:37

Channel Processing Parameters				
Run 5: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	Time Zoned	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	1.25	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.08	ohm.m
Depth Zone Parameters				
Parameter	Value	Start (m)	Stop (m)	
BS	26	906	931	
BS	12.25	931	1906	
All depth are actual.				

Time Zone Parameters

Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
BHK	5.8	16-Dec-2019 08:19:22	16-Dec-2019 11:56:31	22.96	936.19
BHK	6.02	16-Dec-2019 11:56:31	16-Dec-2019 12:02:12	936.19	940.21
BHK	7.52	16-Dec-2019 12:02:12	17-Dec-2019 14:10:12	940.21	1906.17

All depth are at tool zero.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 5

Primary Equipment :

High Pressure without AIM Receiver

AREA

8374

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200	

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



Country.

Norway



Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode